Verified Engineering Blueprints of the RD-0110 Coaxial Swirl Injector

The **RD-0110** (used on the Soyuz-U, Soyuz-FG, and Soyuz-2.1a Block-I upper stage) employs a classic Soviet/Russian liquid–liquid bi-propellant coaxial swirl injector element designed by KBKhA. Each element pairs a **closed-type central Liquid Oxygen (LOX) swirl injector** with a concentric **open-type annular RP-1 Kerosene swirl injector**, Recessed by 1.50 mm to promote internal impingement and supercritical boundary-layer mixing prior to combustion chamber entry.

image_agent_tag_10080197901757997249

image_agent_tag_10080197901757998428

Complete Dimensional Specification Table

All dimensions below are sourced from the published RD-0110 main injector baseline geometry (Huo, Wang, and Yang; Wang, Wang, and Yang, *AIAA Journal*; and *Energies* / *Physics of Fluids* RD-0110 injector studies).

Subsystem	Geometric Parameter	Symbol	Baseline Value	Notes
Overall Body	Total Injector Length	L_{total}	25.80 mm	Includes 1.60 mm rear end-cap
Overall Body	Internal Fluid Length	L_{int}	24.20 mm	Exit plane to rear chamber wall
Overall Body	Rear Cap Wall Thickness	t_{cap}	1.60 mm	Closes upstream LOX swirl chamber
Overall Body	Main Body Outer Diameter	D_{body}	13.00 mm	Radius 6.50 mm on front/rear barrels

Overall Body	Central Flange Diameter	D_{flange}	16.00 mm	Radius 8.00 mm from $x = 7.10$ to 13.80 mm
LOX (Inner)	Swirl Chamber Diameter	$D_{s,o}$	9.00 mm	Inner radius $R_{s,o} = 4.50$ mm
LOX (Inner)	Swirl Chamber Length	$L_{s,o}$	10.40 mm	Spans $x = 13.80$ mm to 24.20 mm
LOX (Inner)	Convergent Cone Axial Length	$L_{c,o}$	1.80 mm	45° contraction from $\varnothing 9.0$ to $\varnothing 5.4$ mm
LOX (Inner)	Nozzle Inner Diameter	$D_{n,o}$	5.40 mm	Radius 2.70 mm (5.10 mm in variant B)
LOX (Inner)	Nozzle Straight Length	$L_{n,o}$	10.50 mm	Spans $x = 1.50$ mm to 12.00 mm
LOX (Inner)	Oxidizer Post Outer Diameter	$D_{p,o}$	7.00 mm	Outer radius 3.50 mm

LOX (Inner)	Oxidizer Post Wall Thickness	h_p	0.80 mm	0.95 mm in the $\varnothing 5.10$ mm bore variant
LOX (Inner)	Post Tip Recess Length	L_r	1.50 mm	Recessed upstream of $x = 0.00$ mm exit
LOX (Inner)	Tangential Inlet Holes	$N_o \times d_{in,o}$	$6 \times \varnothing 1.70$ mm	Evenly spaced every 60° ($r_{in,o} = 0.85$ mm)
LOX (Inner)	Tangential Offset Radius	$R_{in,o}$	3.50 mm	Centerline offset from injector axis
LOX (Inner)	Tangential Port Axial Pos.	$x_{in,o}$	22.50 mm	Measured from downstream exit face
RP-1 (Outer)	Fuel Nozzle Inner Diameter	$D_{s,f}$	10.00 mm	Inner radius $R_{s,f} = 5.00$ mm
RP-1 (Outer)	Fuel Annular Slot Width	δ_f	1.50 mm	Radial gap between $\varnothing 7.0$ and $\varnothing 10.0$ mm

RP-1 (Outer)	Fuel Chamber Axial Depth	$L_{s,f}$	10.50 mm	From exit plane ($x = 0$) to fuel back wall
RP-1 (Outer)	Tangential Inlet Holes	$N_f \times d_{in,f}$	6×0.70 mm	Evenly spaced every 60° ($r_{in,f} = 0.35$ mm)
RP-1 (Outer)	Tangential Offset Radius	$R_{in,f}$	4.50 mm	Centerline offset from injector axis
RP-1 (Outer)	Tangential Port Axial Pos.	$x_{in,f}$	8.80 mm	Measured from downstream exit face

Based on Bazarov's classical hydrodynamic theory of pressure-swirl atomizers, these dimensions yield a geometric swirl characteristic number of:

$$A_o = \frac{R_{in,o} R_{n,o}}{N_o r_{in,o}^2} = \frac{3.50 \times 2.70}{6 \times 0.85^2} = 2.18 (LOXInnerStage)$$

$$A_f = \frac{R_{in,f} R_{s,f}}{N_f r_{in,f}^2} = \frac{4.50 \times 5.00}{6 \times 0.35^2} = 30.61 (RP-1OuterStage)$$

Exact Coordinates & Step-by-Step Build Guide for SolidWorks

You can build the 100% parametric native SolidWorks part in under three minutes using **three features** (**Revolve1**, **Cut-Extrude1** with **CirPattern1**, and **Cut-Extrude2** with **CirPattern2**). Set your SolidWorks document units to **MMGS (Millimeter, Gram, Second)** before starting.

Step 1: Create the Coaxial Body of Revolution (Revolve1)

1. Open a new sketch on the **Front Plane**.
2. Draw a horizontal **Centerline** along the X-axis from $X = 0.00mm$ (downstream injector exit plane) to $X = 25.80mm$ (upstream head face) at $R = 0.00mm$.
3. Connect the following **16 closed-loop vertices** in sequential order using the **Line** tool to form the exact upper cross-sectional profile:

Vertex	Axial Coord. X (mm)	Radial Coord. R (mm)	Geometric Feature Represented
V1	24.20	0.00	Center of LOX swirl chamber rear internal wall
V2	25.80	0.00	Center of external upstream end-cap face
V3	25.80	6.50	Outer top-right corner of rear barrel ($\varnothing 13.00$ mm)
V4	13.80	6.50	Rear step shoulder of central mounting flange
V5	13.80	8.00	Rear outer corner of central mounting flange ($\varnothing 16.00$ mm)
V6	7.10	8.00	Front outer corner of central mounting flange ($\varnothing 16.00$ mm)
V7	7.10	6.50	Front step shoulder of central mounting flange
V8	0.00	6.50	Outer bottom-left corner at injector exit face ($\varnothing 13.00$ mm)
V9	0.00	5.00	Inner exit lip of fuel nozzle ($\varnothing 10.00$ mm)

V10	10.50	5.00	Outer top corner of annular fuel swirl chamber
V11	10.50	3.50	Root of oxidizer post inside fuel swirl chamber ($\varnothing 7.00$ mm)
V12	1.50	3.50	Outer tip corner of recessed oxidizer post ($L_r = 1.50$ mm)
V13	1.50	2.70	Inner tip corner of recessed oxidizer post ($\varnothing 5.40$ mm)
V14	12.00	2.70	Downstream end of 45° LOX convergent nozzle cone
V15	13.80	4.50	Upstream start of 45° LOX convergent cone ($\varnothing 9.00$ mm)
V16	24.20	4.50	Rear internal corner of LOX swirl chamber (close to **V1**)

4. Click **Features > Revolved Boss/Base**, select the horizontal centerline ($R = 0$) as the Axis of Revolution, and revolve 360°.

Step 2: Drill the 6 Tangential LOX Swirl Inlets (Cut-Extrude1 & CirPattern1)

1. Create a **Reference Plane** parallel to the **Top Plane**, offset upward by $3.50mm$ (or sketch directly on the **Front Plane**).
2. The simplest method is to open a sketch directly on the **Front Plane**:
   - Draw a circle with center at $X = 22.50mm$, $Y = 3.50mm$ (the tangential offset radius $R_{in,o} = 3.50mm$), and set its diameter to $1.70mm$ ($r = 0.85mm$).
   - Click **Features > Extruded Cut**, set Direction 1 to **Blind** with depth $7.00mm$ (extending from the Front Plane $Z = 0$ outward through the outer barrel wall at $R = 6.50mm$ in one tangential direction only).

3. Click **Features > Circular Pattern**:
    - Select the central injector axis as the Pattern Axis.
    - Set Angle to 360°, Number of Instances to 6 (Equal Spacing), and select **Cut-Extrude1**.

Step 3: Drill the 6 Tangential RP-1 Fuel Swirl Inlets (Cut-Extrude2 & CirPattern2)

1. Open a new sketch on the **Front Plane**:
    - Draw a circle with center at $X = 8.80mm$, $Y = 4.50mm$ (the fuel tangential offset radius $R_{in,f} = 4.50mm$), and set its diameter to $0.70mm$ ($r = 0.35mm$).
    - Click **Features > Extruded Cut**, set Direction 1 to **Blind** with depth $8.50mm$ (cutting tangentially from the fuel swirl chamber at $R = 5.00mm$ out through the $\varnothing 16.00\text{ mm}$ central flange).
2. Click **Features > Circular Pattern**:
    - Select the central injector axis as the Pattern Axis.
    - Set Angle to 360°, Number of Instances to 6 (Equal Spacing), and select **Cut-Extrude2**.
    - *(Note: Verify whether your test configuration requires co-swirling or counter-swirling propellants; in standard baseline RD-0110 co-swirl studies, both Direction 1 cuts point in the same $+Z$ tangential orientation).*